TECHNISCHE UNIVERSITÄT BERLIN

Scoring Rules for Probabilistic Forecasting

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[1]Mathematics
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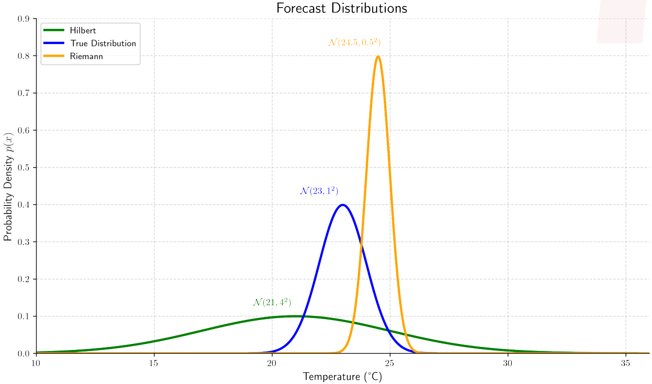
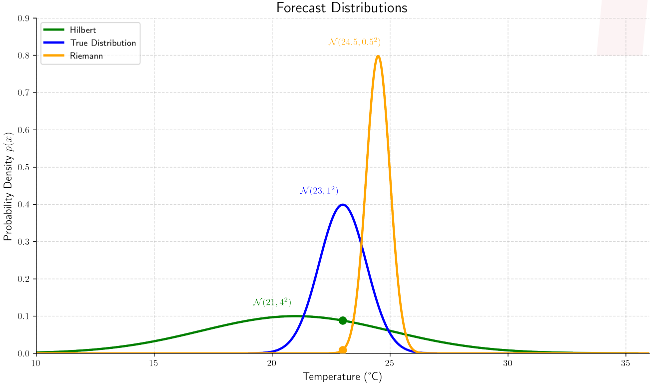
Motivation



$$S(P, \omega) := \log(p(\omega))$$

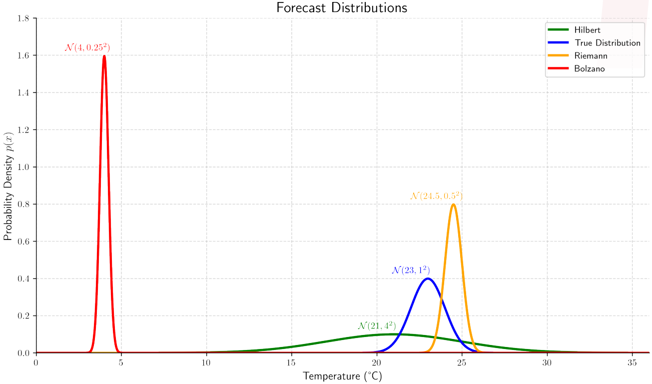
Motivation



Hilbert: $S(P_H, 23) = \log(p_H(23)) \approx -2.4$

Riemann: $S(P_R, 23) = \log(p_R(23)) \approx -4.7$

Motivation



Hilbert: $S(P_H, 4) = \log(p_H(4)) \approx -11.3$

Riemann: $S(P_R, 4) = \log(p_R(4)) \approx -840.7$

Bolzano: $S(P_B, 4) = \log(p_B(4)) \approx -0.2$

Expected Score

Definition

Let Q be the true distribution, then

$$S(P, Q) = \mathbb{E}^{Q}\left[S(P, \omega)\right]$$

is called the **Expected Score**.

Properness

Fourier knows the truth Q. Should he also forecast Q?

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Fourier knows the truth Q. Should he also forecast Q?

Definition

A scoring rule is called (strictly) **proper**, if

$$S(Q, Q) \geq S(P, Q)$$

for all P.

Improper Scoring Rule

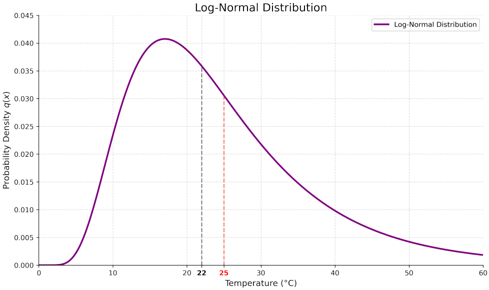
Linear Score

Define

$$S(P, \omega) := |\omega - \mathrm{median}(P)|$$

Let Q be Log-Normal. Fourier would rather predict $P \sim \mathcal{N}(23, \sigma^2)$. Then S is improper

$$S(P, Q) > S(Q, Q).$$



Mathematical Characterization

Definition

- Ω sample space
- $\mathcal{A}$ σ-Algebra on subsets of Ω
- $\mathcal{P}$ convex class of probability measures
- $P \in \mathcal{P}$ probabilistic forecast

$S : \mathcal{P} \times \Omega \to \overline{\mathbb{R}} :\ S(P, \cdot)$ quasi-integrable $\forall P \in \mathcal{P}$ is called a **Scoring Rule**. In the event ω the probabilistic forecast gives the **reward** $S(P, \omega)$.

Mathematical Characterization

Definition

- For $P, Q \in \mathcal{P}$ we define the **Expected Score**

$$S(P,Q) = \mathbb{E}^Q\left[S(P,\omega)\right] = \int_\Omega S(P,\omega)\mathrm{d}Q(\omega).$$

- S is called **regular** relative to $\mathcal{P}$ if

$$S(P,Q) \in [-\infty, \infty)$$

for all $P, Q \in \mathcal{P}$ where $P \neq Q$ and $S(P,P) \in \mathbb{R}$ for all $P \in \mathcal{P}$.

Mathematical Characterization

Definition

- A scoring rule is called **proper** if

$$S(Q, Q) \geq S(P, Q) \qquad \forall P, Q \in \mathcal{P}.$$

- A scoring rule is called **strictly proper** if

$$S(Q, Q) > S(P, Q) \qquad \forall P, Q \in \mathcal{P},\ P \neq Q.$$

- Proper rules encourage honesty.

Divergences

Definition

Let S be proper relative to $\mathcal{P}$. The function

$$G : \mathcal{P} \to \mathbb{R}, \quad G(P) = \sup_{Q \in \mathcal{P}} S(Q, P)$$

is called the **information measure** or (generalized) **entropy function**.

Note: G is *convex* as $\sup$ over affine functions.

Convexity

Definition

A function $G : \mathcal{P} \to \mathbb{R}$ is **convex** if

$$G\left((1-\lambda)P + \lambda Q\right) \leq (1-\lambda)G(P) + \lambda G(Q)$$

for all $\lambda \in (0,1)$ and $P, Q \in \mathcal{P}$.

Divergence

Definition

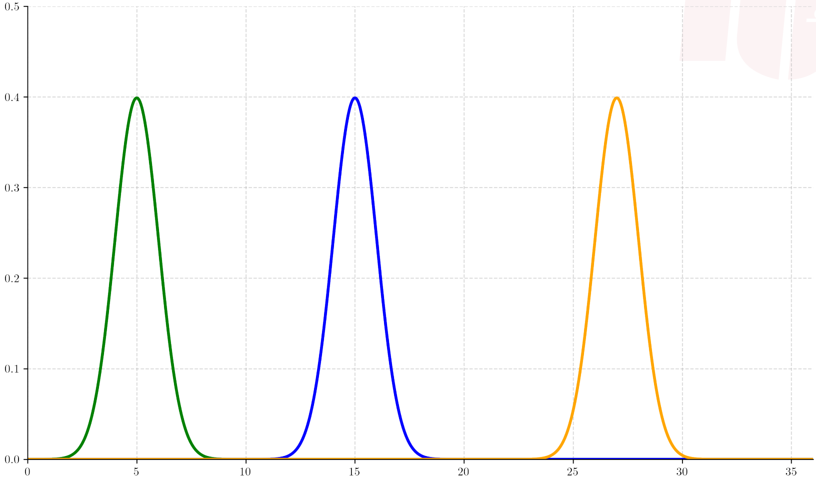
- For a regular and proper scoring rule S we call

$$d : \mathcal{P}^2 \to [0, \infty), \quad d(P, Q) = S(Q, Q) - S(P, Q)$$

  the **divergence function**.
- If Ω is finite, G sufficiently smooth, then d is the **Bregman divergence** associated with the (convex) entropy function G.

Note: If S is strictly proper, $d(P, Q) > 0$ for $P \neq Q$.

Bregman Divergence



Categorical Scoring Rules

Setting

- Let $\Omega = \{1, ..., n\}$ and $\mathcal{P} = \Delta_n = \left\{p \in \mathbb{R}^n : p \geq 0, \sum_{j=1}^n p_j = 1\right\}$.
- $S(\cdot, i) : \mathcal{P} \to \overline{\mathbb{R}}$.

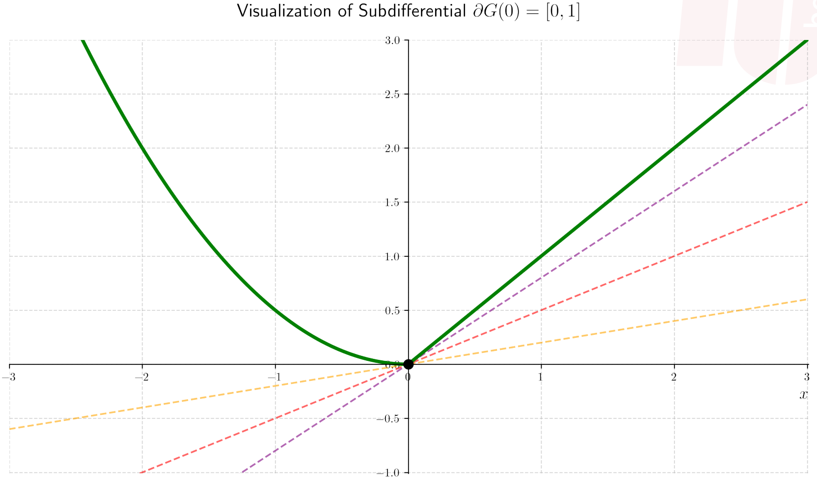
Definition

Let $G : \mathcal{P} \to \mathbb{R}$ be convex. For $p \in \mathcal{P}$ the **subdifferential** of G at p is defined as

$$\partial G(p) := \{v \in \mathbb{R}^n : G(q) \geq G(p) + \langle v, q - p \rangle \text{ for all } q \in \Delta_n\}.$$

A $v \in \partial G(p)$ is called a **subgradient**. Note: a subgradient is a vector in $\mathbb{R}^n$, not necessarily a probability vector.

Subdifferential



Categorical Scoring Rules

Theorem (McCarthy, Savage)

A scoring rule S is proper if and only if $G(p) = S(p,p)$ is convex on $\mathcal{P}$ and $\sum_{i=1}^{n} S(p,i)e_i \in \partial G(p)$ for all $p \in \mathcal{P}$.

Corollary

Every bounded convex entropy function G on $\mathcal{P}$ generates a regular proper scoring rule.

Brier Score (Quadratic Score)

Entropy Function

$$G(p) = \sum_{j=1}^{n} p_j^2 - 1$$

Scoring Rule

$$S(p, i) = -\sum_{j=1}^{n} \left(\delta_{i,j} - p_j\right)^2 = 2p_i - \sum_{j=1}^{n} p_j^2 - 1$$

Brier Score (Quadratic Score)

Entropy Function

$$G(p) = \sum_{j=1}^{n} p_j^2 - 1$$

Scoring Rule

$$S(p,i) = -\sum_{j=1}^{n} (\delta_{i,j} - p_j)^2 = 2p_i - \sum_{j=1}^{n} p_j^2 - 1$$

Bregman Divergence

$$d(p,q) = \sum_{j=1}^{n} (p_j - q_j)^2$$

Logarithmic Score

Entropy Function (Negative Shannon Entropy)

$$G(p) = \sum_{j=1}^{n} p_j \log(p_j)$$

Scoring Rule

$$S(p, i) = \log(p_i)$$

Logarithmic Score

Entropy Function (Negative Shannon Entropy)

$$G(p) = \sum_{j=1}^{n} p_j \log(p_j)$$

Scoring Rule

$$S(p, i) = \log(p_i)$$

Bregman Divergence (Kullback-Leibler Divergence)

$$d(p, q) = \sum_{j=1}^{n} q_j \log\left(\frac{q_j}{p_j}\right)$$

Zero-One Score

Scoring Rule

$$S(p,i) = \begin{cases} 1/\#M(p) & \text{if } i \in M(p) \\ 0 & \text{else} \end{cases}$$

where

$$M(p) = \underset{1 \leq j \leq n}{\arg\max}\; p_j$$

are the modes of p.

Entropy Function

$$G(p) = \max_{1 \leq j \leq n} p_j$$

Zero-One Score

Scoring Rule

$$S(p,i) = \begin{cases} 1/\#M(p) & \text{if } i \in M(p) \\ 0 & \text{else} \end{cases}$$

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are the modes of p.

Entropy Function

$$G(p) = \max_{1 \le j \le n} p_j$$

Divergence Function

$$d(p,q) = \max_{1 \le j \le n} q_j - \frac{\sum_{j \in M(p)} q_j}{\#M(p)}$$

Zero-One Score

Note: This is not a Bregman divergence, since G is not differentiable and not strictly convex.

Scoring Rules for Continuous Variables

Goal for this section:

- Define Scoring Rules for continuous variables
- See what the differences between them are
- Consider, what may be benefits or drawbacks of certain Scoring Rules

Scoring Rules for Density Forecasts

Setting of the Forecasts:

- μ be σ-finite on $(\Omega, \mathcal{A})$
- $\mathcal{L}_\alpha$ probability measures absolutely continuous with respect to μ, $\alpha > 1$
- Each measure from $\mathcal{L}_\alpha$ has μ-density p with

$$\|p\|_\alpha = \left(\int p(\omega)^\alpha \mu(d\omega) \right)^{\frac{1}{\alpha}} < \infty$$

- p will be called density forecast

Note: p is not uniquely defined for some measure $P \in \mathcal{L}_\alpha$

Examples for Scoring Rules for Density Forecasts

- Quadratic Score:

$$\mathsf{QS}(p,\omega) = 2p(\omega) - \|p\|_2^2$$

- Strictly proper relative to $\mathcal{L}_2$
- Expected score:

$$G(p) = \|p\|_2^2$$

Examples for Scoring Rules for Density Forecasts

- Pseudospherical Score:

$$\mathsf{PseudoS}(p,\omega) = \frac{p(\omega)^{\alpha-1}}{\|p\|_\alpha^{\alpha-1}}$$

- Strictly proper relative to class $\mathcal{L}_\alpha$

The Logarithmic Score

Definition:

$$\mathsf{LogS}(p, \omega) = \log p(\omega)$$

- Strictly proper relative to $\mathcal{L}_1$ of dominated measures
- Widely used under various names
- Under certain conditions, all proper scoring rules that depend on p only through the value $p(\omega)$ are equivalent

Using the Logarithmic Score (PASCAL Challenges)

Setting (Regression task):

- Goal is to give a prediction for the correlation between variables
- Predictions given in form of density function
- Scoring of the prediction to determine a winner

Scoring the Predictive Density

$$L = \frac{1}{n} \sum_{i=1}^{n} \log p(y_i = t_i \mid x_i)$$

Possible Problems of the Logarithmic Score

Expected point masses can dominate the score:

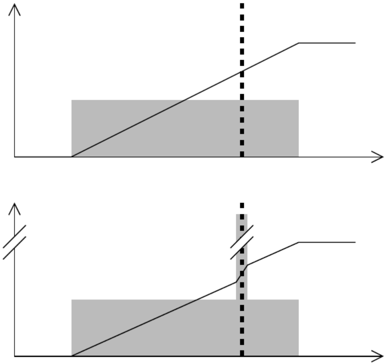

Fig. 1: Figure 8 of [3] showing the issue with the logarithmic score

Possible Problems with this Scoring Function

- Setting artificially high densities could get rewarded
- Maybe looking at distributions functions could help

Scoring for Distribution Functions and CRPS

Scoring for Distribution Functions

Motivation for looking at distribution functions instead of densities:

1. Solving the point mass problem of logarithmic score
2. Allowing samples to easily represent distribution functions:

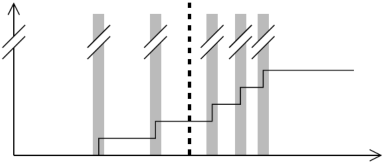

Fig. 2: Figure 9 of [3] about a way of representing a distribution function

3. Considering "distance"

Continuous Ranked Probability Score (CRPS)

Definition: $\mathcal{P}$ set of Borel probability measures on $\mathbb{R}$.

$$\mathsf{CRPS}(F, x) = -\int_{-\infty}^{\infty} (F(y) - \mathbb{1}\{y \geq x\})^2 dy$$

Continuous Ranked Probability Score (CRPS)

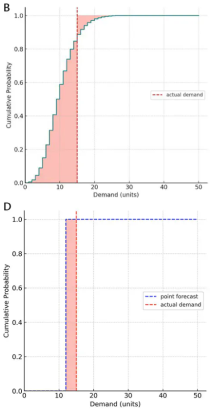

Fig. 3: Visualisation of CRPS from
https://www.lokad.com/continuous-ranked-probability-score/

Step by Step Visualisation

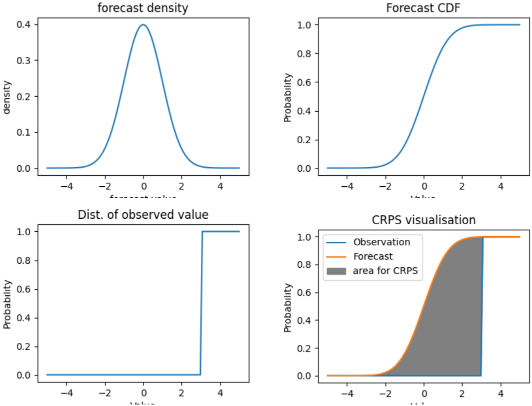

Fig. 4: Visualisation of the CRPS for predicted normal dist.

Properties of the CRPS

- Proper relative to $\mathcal{P}$
- Strictly proper relative to subclass $\mathcal{P}_1$ of prob. measures with finite first moments
- Divergence function

$$d(F,G) = \int_{-\infty}^{\infty} (F(y) - G(y))^2 dy$$

- Some sort of sensitivity for "distance"

Properties of the CRPS

Alternative Definition:

$$\mathrm{CRPS}(F, x) = \frac{1}{2} E_F[|X - X'|] - E_F[|X - x|]$$

- X, X' independent copies of a random variable with dist. F
- Definition will come up again later

Possible Downsides of the CRPS

- The integral is not always easy to compute
- One probably has to use numerical methods such as quadrature formulas:

$$S = \sum_i (x_{i-1} - x_i) \cdot \frac{(F(x_{i-1}) - D(x_{i-1})^2 + (F(x_i) - D(x_i))^2}{2}$$

Possible other Scoring Functions

- Generalization of CRPS on $\mathbb{R}^m$ (energy score)
- Scoring rules only depending on first and second moment

Kernel Scores

Is there a general way to describe or define Scoring Rules?

Kernel Scores

Definition (kernel): Let Ω be a nonempty set. A function $g\colon \Omega \times \Omega \to \mathbb{R}$ is called a negative definite kernel if it is symmetric in its arguments and

$$\sum_{i=1}^{n} \sum_{j=1}^{n} a_i a_j g(x_i, x_j) \leq 0$$

for all positive integers n and all $a_1, \ldots, a_n$ that sum to 0

Kernel Scores

Theorem (Theorem 4 of [2]): Let Ω be a Hausdorff space (a space where distinct points have disjoint neighborhoods around them) and let $g\colon \Omega \times \Omega \to \mathbb{R}$ be a non-negative, continuous and negative definite kernel. Also let P be a Borel probability measure on Ω. Let X, X' be independent with distribution P. In this case the scoring rule

$$S(P, x) = \frac{1}{2} E_P[g(X, X')] - E_P[g(X, x)]$$

is proper regarding the class of Borel probability measures on Ω

CRPS as Kernel Score

- Take $g(x, x') := |x - x'|$ as a kernel in our sense.
- The theorem then gives a score

$$S(P, x) = \frac{1}{2} E_P[|X - X'|] - E_P[|X - x|]$$

Brier score

We look at the following scenario:

- $\Omega = \{0, 1\}$
- $g(0,0) = g(1,1) = 0$ and $g(1,0) = g(0,1) = 1$
- p_0, p_1 as probabilities of each event 1 or 0

Then:

$$S(P, 1) = \frac{1}{2}(p_0 p_1 + p_1 p_0) - p_0 = p_0 \cdot (1 - p_0) - p_0 = -p_0^2$$

Brier Score as Kernel Score

Scoring rule	*S(p, 1)*	*S(p, 0)*
Brier	$-(1-p)^2$	$-p^2$

Fig. 5: Part of the Table in [2] about Scores on binary forecasts

Complex Kernel Scores

We can allow complex-valued kernels h as follows:

- $h\colon \Omega \times \Omega \to \mathbb{C}$ and $h(x, x') = \overline{h(x', x)}$
- h positive definite:

$$\sum_{i=1}^{n} \sum_{j=1}^{n} c_i \bar{c}_j h(x_i, x_j) \geq 0$$

Complex Kernel Scores

General Idea: Generalizing the theorem for this case. By doing that we can can look at $\mathbb{R}^m$ and choose h the right way to get

$$S(P, x) = -|\phi_P(y) - e^{i\langle x,y\rangle}|^2.$$

Here ϕ_P is the characteristic function of the predictive distribution.

Scoring Rules for Quantiles

- Instead of predictive distributions we look at sections of the distribution
- The predictions now are the points at the edges of these sections

Motivation for Quantile Scores (Value at Risk)

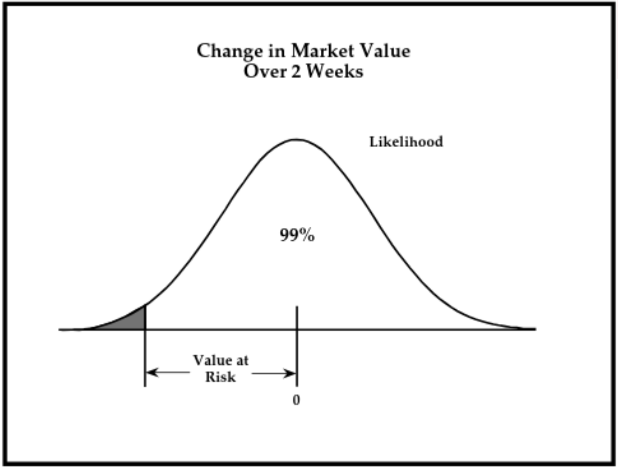

Fig. 6:Figure 1 of [1] for a intuition behind the VaR

Intuition behind Quantiles

Setting: We want to have the values $r_1, \ldots, r_k$ at given probability levels $\alpha_1, \ldots, \alpha_k \in (0,1)$ such that

$$P(x \leq r_k) = \alpha_k.$$

We are trying to score the predicted values r_k.

Proper Scoring Rules for Quantiles

- Let $q_1, \dots, q_k$ denote the true quantiles of distribution P
- Scoring rules for quantiles will have the form $S(r_1, \dots, r_k, x)$ with

$$S(r_1, \dots, r_k, P) = \int S(r_1, \dots, r_k, x) dP(x).$$

- A Scoring Rule is called proper if

$$S(q_1, \dots, q_k, P) \geq S(r_1, \dots, r_k, P)$$

for all $r_1, \dots, r_k \in \mathbb{R}$ and probability measures P.

Proper Scoring Rules for Quantiles

Theorem: Let s be a nondecreasing function and h an arbitrary function, then

$$S(r,x) = \alpha s(r) + (s(x) - s(r))\mathbb{1}_{\{x \leq r\}} + h(x)$$

is proper for predicting a single quantile at level $\alpha \in (0,1)$.

Scoring Multiple Quantiles

- Choose $s_i, i = 1, \ldots, k$ as in the theorem and
- h arbitrary
- The sum over all scores $i = 1, \ldots, k$ is proper for scoring k quantiles

Example

Set $s(x) = x, h(x) = -\alpha x$. Then we get the Score

$$S(r, x) = \alpha r + (x - r)\mathbb{1}_{\{x \leq r\}} - \alpha x = (x - r)(\mathbb{1}_{\{x \leq r\}} - \alpha)$$

Special Case: Interval Score

- Prediction of quantiles at $\frac{\alpha}{2}, 1 - \frac{\alpha}{2}$
- E.g. prediction of central 50% by using $\alpha = \frac{1}{2}$
- Can be scored like quantiles

Potential Further Applications

- Use Scoring Rules to score forecasting models
- Let $X = (X_1, \ldots, X_n)$
- Forecast of a model determined up to a parameter θ

Scoring Function in this Applications

Comparing Forecasting Models:

- Basic score:

$$S_{k,B} = \sum_{t=1}^{n} \log p(X_t \mid X^{t-1}, H_k)$$

Unordered Data

If the observed data does not arrive in a certain order we look at random samples with random uniformly distributed size. The modification of the score from before opens up the possibility of cross-validation

References

[1] Darrell Duffie and Jun Pan. An overview of value at risk. *Journal of Derivatives*, 4:7–49, 1997.

[2] Tilmann Gneiting and Adrian Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102:359–378, 03 2007.

[3] Jukka Kohonen and Jukka Suomela. *Lessons Learned in the Challenge: Making Predictions and Scoring Them*, volume 3944, pages 95–116. Springer, 04 2006.

[4] J. Quiñonero-Candela, C. E. Rasmussen, F. Sinz, O. Bousquet, and B. Schölkopf. Evaluating predictive uncertainty challenge. *Machine Learning Challenges: Evaluating Predictive Uncertainty, Visual Object Classification and Recognizing Textual Entailment*, pages 1–27, 2005.

[5] David A. Unger. A method to estimate the continuous ranked probability score. *Preprints of the Ninth Conference on Probability and Statistics in Atmospheric Sciences*, pages 206–213, 1985.

Further References

CRPS-Visualization code:
https://scores.readthedocs.io/en/stable/tutorials/CRPS_for_CDFs.html
Figure 3: https://www.lokad.com/continuous-ranked-probability-score/